Zili Tan

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Research Interests

Self-Improving AI Systems, Learning Dynamics, Continual Adaptation, and Multi-Agent Systems.

Academic Background

University of Amsterdam & Vrije Universiteit Amsterdam *09/2025 – 06/2027 (expected)*
MSc Physics and Astronomy (joint degree), Theoretical Physics track Current average grade: 8.62/10

Sun Yat-sen University *09/2021 – 07/2025*
Bachelor of Science in Physics Average grade: 8.57/10

Publications & Preprints

Meng Bi, Jinhai Yang, **Zili Tan**, Bowei He, Hong Huang, and Xue Liu. *Wright–Gödel Machine: Mitigating Homogenization in Self-Improving Agents.* **Under review at ICLR 2027.**

Research Experience

Continual Learning in Resource-Constrained Agents *10/2026 – Present*
Exploratory research

- Studying whether experience remains useful across changing task streams when agent memory and compute are limited.
- Built and locally tested a research prototype and prepared a manuscript draft.
- Designed controlled evaluations under matched resource limits to measure cumulative task performance and how prior experience helps with recurring and held-out tasks.

Agentic AI Workflows for SMEFT Event Generation at the LHC *10/2026 – Present*
MSc Thesis, Advisor: Prof. Wouter Verkerke, Nikhef / University of Amsterdam

- Building the simulation and validation foundation for an agentic SMEFT event-generation workflow for LHC analyses.
- Generated and analyzed a 2,000-event SMEFTsim Higgs-decay sample with 11 additional event weights and 48 ROOT histograms.
- Checked event-count consistency, stable-particle momentum preservation, and same-seed repeatability in a 200-event MadGraph–Pythia–HepMC3–Delphes/Parnassus pipeline.
- Published bilingual SMEFTsim and detector-simulation tutorials covering isolated Linux environments, manual simulation workflows, and ROOT analysis.
- Learning about agent harnesses and LangGraph for workflow orchestration.

Genetic Homogenization in Self-Improving Coding Agents *08/2026 – 09/2026*
Collaborative project with Meng Bi et al.

- Investigated diversity collapse in recursively self-improving coding agents across Darwin, Huxley, and Mendel Gödel Machine variants.
- Co-introduced genetic homogenization to characterize the loss of heritable design diversity during iterative self-improvement.
- Studied the exploration–exploitation tradeoff underlying this collapse and proposed EXP3 as an alternative to Thompson Sampling for maintaining architectural diversity.
- Co-introduced an evaluation strategy inspired by BAI-MCTS.

Observer-Relative Inflationary Histories in de Sitter JT Gravity *06/2026 – 09/2026*
Advisors: Prof. Ben Freivogel and Prof. Jan Pieter van der Schaar, University of Amsterdam

- Studied de Sitter JT (Jackiw–Teitelboim) gravity coupled to a homogeneous inflaton and a weakly coupled observer.
- Found that the current observer-relative gravitational path integral (GPI) prescription does not normalize the no-boundary state, indicating a possible inconsistency of the Hartle–Hawking proposal in the observer-relative Hilbert space.

Lorentzian Gravitational Path Integral in the de Sitter Static Patch *04/2026 – 06/2026*
Advisor: Prof. Jan Pieter van der Schaar, University of Amsterdam

- Surveyed how the Lorentzian path-integral approaches can avoid the conformal factor problem.
- Used Claude Code to expand the Einstein–Hilbert action around the D-dimensional de Sitter static patch.
- Obtained the one-loop spectrum of de Sitter space from the Lorentzian gravitational path integral.

Holographic Cosmology and Inflationary Cosmology *10/2022 – 06/2025*
BSc Thesis, Advisor: Prof. Chengcheng Han, Sun Yat-sen University

- Studied the standard cosmological model and inflationary cosmology in Professor Han's Research Group.
- Reviewed the holographic principle, holographic entropy bounds, and models proposed by Banks and Fischler.
- Analyzed how holographic models address the low-entropy initial condition problem and evaluated the prospects for testing holographic cosmology with future CMB and primordial gravitational-wave observations.

Competitions & Awards

National Training Program of Innovation for Undergraduates *12/2022 – 03/2024*
Presenter and Team Member

- Reviewed slow-roll models and compared their predicted power spectra with Planck TT observations.
- The final report was rated **Excellent** (03/2024) in the National Training Program of Innovation for Undergraduates.

China Undergraduate Physics Tournament (CUPT) *12/2021 – 10/2022*
Contestant representing SYSU

- Conducted literature reviews on fluid mechanics and Hamiltonian mechanics; designed experiments and built theoretical models for several real-life-inspired physics problems; presented results in the Central South Regional and National rounds.
- Awarded the **National First Prize** (10/2022) of CUPT.
- Awarded the **First Prize** in the Central South Regional Competition (06/2022).

Extracurricular Activities

SYSU Freshman Mentoring Program *09/2024 – 09/2025*
Mentor

- Guided a group of incoming students throughout their first year, helping them adapt to university life and answering their questions about study and campus life.

SYSU Torchwood Association of Physics *10/2022 – 10/2024*
Governing Member

- Organized information-sharing seminars and physics contests, including the CUPT Simulation Game.
- Served as a judge in several contests.

Skills

- Python, Mathematica, LaTeX, and Linux.